

Step-by-Step Guide: Generating and Analyzing System Logs

This guide explains how to generate, extract, and analyze your own system logs using the AI Security Analyst project.

Step 1: Generating System Logs

System and application logs are typically generated automatically based on your configuration.

- **Nginx/Apache:** Web servers automatically generate access logs. Nginx logs are usually at `/var/log/nginx/access.log`, and Apache at `/var/log/apache2/access.log`.
- **Custom Applications:** Use standard logging libraries (like Python's logging or Node.js's winston) to write application events to a file (e.g., `app.log`).
- **Windows IIS:** Logs are typically located in `C:\inetpub\logs\LogFiles`.

Step 2: Extracting the Logs

Once the logs are generated, you need to extract them to a location where you can upload them to the application.

- **Linux/macOS:** Open a terminal and copy the log to a user directory: `cp /var/log/nginx/access.log ~/Desktop/my_access.log`.
- **Windows:** Navigate to the log directory using File Explorer and copy the `.log` file to your Desktop.
- Ensure you have read permissions for the file.

Step 3: Preparing Logs for Analysis

The AI Security Analyst expects text-based log files (e.g., Common Log Format or JSON). Open your extracted `.log` file in a text editor (like Notepad or VS Code) to verify it contains readable text and no binary corruption.

Step 4: Uploading to AI Security Analyst

- 1 Start the AI Security Analyst application by running both the backend and frontend servers.
- 2 Open the web application in your browser.
- 3 Navigate to the **Log Upload** component.
- 4 Click **Browse** or drag-and-drop your extracted `.log` file into the drop zone.
- 5 Click the **Upload and Analyze** button.

Step 5: Reviewing the Analysis

- 1 Wait for the AI to parse the logs and process them using the vector store.
- 2 View the **Threat Timeline** to see any detected anomalies or attacks (like XSS or SQL Injection).
- 3 Use the **Chat Panel** to ask specific questions about the log file (e.g., 'Were there any unauthorized access attempts?').